

Assignment 4: HTML Basics- Practical Assignment

Objective

Build a complete Personal Portfolio Webpage using HTML concepts learned in class

Part A: Short Theory Questions

1. What does HTML stand for?

HTML stands for **Hyper Text Markup Language**. It is used to create and structure web pages.

2. What is the difference between a Tag and an Element?

A **tag** is the markup inside angle brackets, such as `<p>`. An **element** consists of the opening tag, content, and closing tag, such as

`<p>Hello</p>`.

3. What is an Attribute?

An attribute provides additional information about an HTML element. It is written inside the opening tag, for example,

href in `<a href="page.html">`.

4. What is Semantic HTML?

Semantic HTML uses meaningful tags like

`<header>`

`<nav>`

`<footer>`

to describe the content. It improves readability, accessibility, and SEO.

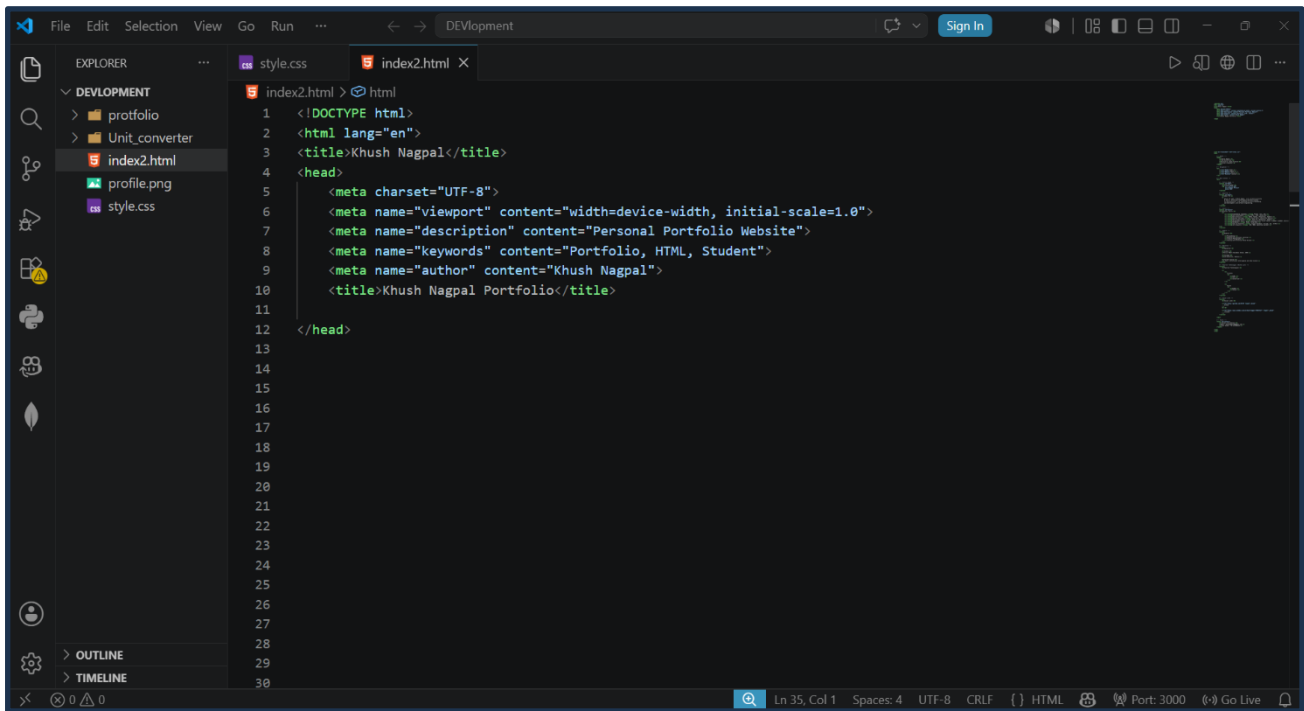
5. Why is the alt attribute important for images?

The alt attribute provides alternative text for an image if it cannot be displayed. It also helps screen readers describe images to visually impaired users.

Part B: Practical Tasks Task

1: Create Basic HTML Structure Create a file named: index.html Requirements:

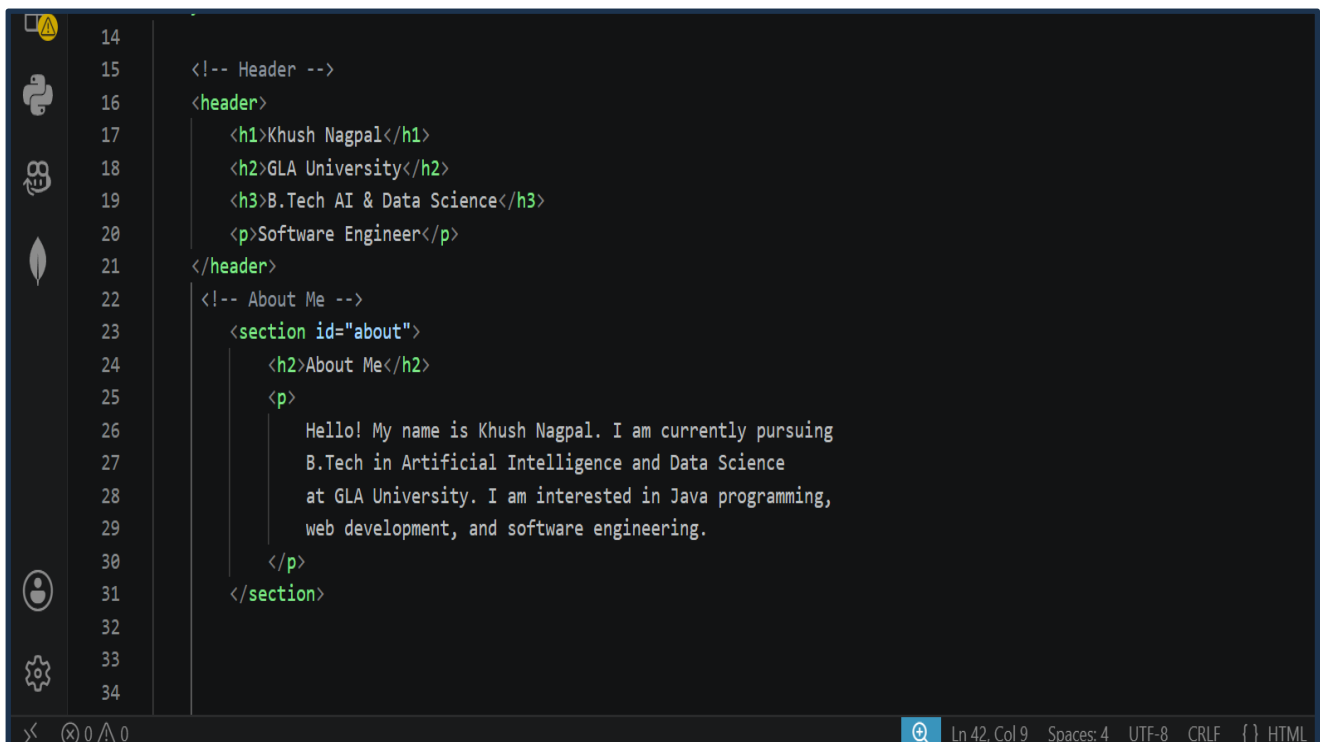
- Use HTML5 Boilerplate
- Add a meaningful title
- Add required meta tags
- Set language to English



```
1 <!DOCTYPE html>
2 <html lang="en">
3 <title>Khush Nagpal</title>
4 <head>
5   <meta charset="UTF-8">
6   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   <meta name="description" content="Personal Portfolio Website">
8   <meta name="keywords" content="Portfolio, HTML, Student">
9   <meta name="author" content="Khush Nagpal">
10  <title>Khush Nagpal Portfolio</title>
11
12 </head>
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
```

Task 2: Personal Information Section Create:

- Your Name using h1
- College Name using h2
- Course Name using h3
- Introduction using p tag



```
14
15 <!-- Header -->
16 <header>
17   <h1>Khush Nagpal</h1>
18   <h2>GLA University</h2>
19   <h3>B.Tech AI & Data Science</h3>
20   <p>Software Engineer</p>
21 </header>
22 <!-- About Me -->
23 <section id="about">
24   <h2>About Me</h2>
25   <p>
26     Hello! My name is Khush Nagpal. I am currently pursuing
27     B.Tech in Artificial Intelligence and Data Science
28     at GLA University. I am interested in Java programming,
29     web development, and software engineering.
30   </p>
31 </section>
32
33
34
```

Task 3: Skills Section Create a section named:

Skills Add an Ordered List containing at least:

- HTML
- CSS
- JavaScript
- React

```
53 <!-- Skills -->
54 <section id="skills">
55 <h2>Technical Skills</h2>
56 <ul>
57 <li><strong>Programming Languages:</strong> Python, Java, SQL</li>
58 <li><strong>Web Technologies:</strong> HTML5, CSS3, JavaScript</li>
59 <li><strong>Data Analysis:</strong> NumPy, Pandas, Matplotlib, Seaborn</li>
60 <li><strong>Machine Learning:</strong> Scikit-learn, TensorFlow (Basics)</li>
61 <li><strong>Data Visualization:</strong> Power BI, Matplotlib, Seaborn</li>
62 <li><strong>Spreadsheet Tools:</strong> Advanced Excel (Pivot Tables, VLOOKUP, XLOOKUP,
Charts)</li>
63 <li><strong>Databases:</strong> MySQL, PostgreSQL</li>
64 <li><strong>Tools & Platforms:</strong> Jupyter Notebook, VS Code, Git, GitHub</li>
65 <li><strong>Core Subjects:</strong> OOP, DBMS, Operating Systems</li>
66 </ul>
67 </section>
68
```

Task 4: Hobbies Section Create a section named: Hobbies

```
70 <section>
71 <h2>Hobbies</h2>
72 <ul>
73 <li>Programming</li>
74 <li>Reading and Continuous Learning</li>
75 <li>Creative Photography</li>
76 <li>Cooking and Perfecting Recipes</li>
77 </ul>
78 </section>
79
```

Task 5: Education Section Create a section showing: • School Name • College Name • Current Course

```
80 <!-- Education -->
81 <section>
82 <h2>Education</h2>
83
84 <h3>School</h3>
85 <p>Police Modern Secondary School, AGRA</p>
86
87 <h3>College</h3>
88 <p>GLA University, Mathura</p>
89
90 <h3>Current Course</h3>
91 <p>B.Tech in Artificial Intelligence and Data Science</p>
92 </section>
93
```

Task 6: Add an Image Add: • One profile image • alt attribute • width attribute

```
33
34      <!-- Profile Image -->
35      <section id="home">
36          <h2>Profile Image</h2>
37          
40      </section>
41
```

Task 7: Add Hyperlinks Create links for: • GitHub • LinkedIn

```
118      <!-- Social Links -->
119      <section>
120          <h2>Social Links</h2>
121
122          <a href="https://github.com/KN-05" target="_blank">
123              GitHub
124          </a>
125          <br><br>
126
127          <a href="https://www.linkedin.com/in/khush-nagpal-576827327/" target="_blank">
128              LinkedIn
129          </a>
130      </section>
```

Task 8: Create Semantic HTML Layout Use the following tags:

```
index2.html > html > body > main > section#home > img
1 <!DOCTYPE html>
2 <html lang="en">
3 <title>Khush Nagpal</title>
4 <head>
5   <meta charset="UTF-8">
6   <meta name="viewport" content="width=device-width, initial-scale=1.0">
7   <meta name="description" content="Personal Portfolio Website">
8   <meta name="keywords" content="Portfolio, HTML, Student">
9   <meta name="author" content="Khush Nagpal">
10  <title>Khush Nagpal Portfolio</title>
11 </head>
12 <body>
13
14   <!-- Header -->
15   <header>
16     <h1>Khush Nagpal</h1>
17     <h2>GLA University</h2>
18     <h3>B.Tech AI & Data Science</h3>
19     <p>Software Engineer</p>
20   </header>
21
22   <!-- Navigation -->
23   <nav>
24     <a href="#home">Home</a> |
25     <a href="#about">About</a> |
26     <a href="#skills">Skills</a> |
27     <a href="#contact">Contact</a>
28   </nav>
29
30   <!-- Main Content -->
31   <main>
32
33     <!-- Profile Image -->
34     <section id="home">
35       <h2>Profile Image</h2>
36       
41     <section>
42       <h2>Hobbies</h2>
43       <ul>
44         <li>Programming</li>
45         <li>Reading and Continuous Learning</li>
46         <li>Creative Photography</li>
47         <li>Cooking and Perfecting Recipes</li>
48       </ul>
49     </section>
50
51     <!-- Education -->
52     <section>
53       <h2>Education</h2>
54
55       <h3>School</h3>
56       <p>Police Modern Secondary School, AGRA</p>
57
58       <h3>College</h3>
59       <p>GLA University, Mathura</p>
60
61       <h3>Current Course</h3>
62       <p>B.Tech in Artificial Intelligence and Data Science</p>
63     </section>
64
65     <!-- Favorite Technologies (Nested List) -->
66     <section>
67       <h2>Favorite Technologies</h2>
68
69       <ul>
70         <li>Frontend
71           <ul>
72             <li>HTML</li>
73
74
75       </ul>
76     </section>
77
78   </main>
79
80   <!-- Footer -->
81   <footer id="contact">
82     <h2>Contact Information</h2>
83     <p>Email: khushnagpal792@gmail.com</p>
84     <p>Phone Number: +91 8979802635</p>
85   </footer>
86
87 </body>
88 </html>
```

Task 9: Mini Portfolio Project Combine all previous tasks into a single webpage.

Khush Nagpal

B.Tech AI & Data Science
Software Engineer

[Home](#) | [About](#) | [Skills](#) | [Contact](#) c

Profile Image



About Me

Hello! My name is Khush Nagpal. I am currently pursuing B.Tech in Artificial Intelligence and Data Science at GLA University. I am interested in Java programming, web development, and software engineering.

Technical Skills

- **Programming Languages:** Python, Java, SQL
- **Web Technologies:** HTML5, CSS3, JavaScript
- **Data Analysis:** NumPy, Pandas, Matplotlib, Seaborn
- **Machine Learning:** Scikit-learn, TensorFlow (Basics)
- **Data Visualization:** Power BI, Matplotlib, Seaborn
- **Spreadsheet Tools:** Advanced Excel (Pivot Tables, VLOOKUP, XLOOKUP, Charts)
- **Databases:** MySQL, PostgreSQL
- **Tools & Platforms:** Jupyter Notebook, VS Code, Git, GitHub
- **Core Subjects:** OOP, DBMS, Operating Systems

Hobbies

- Programming
- Reading and Continuous Learning
- Creative Photography
- Cooking and Perfecting Recipes

Education

School
Police Modern Secondary School, AGRA

College
GLA University, Mathura

Current Course
B.Tech in Artificial Intelligence and Data Science

Favorite Technologies

- Frontend
 - HTML
 - CSS
 - JavaScript
- Python
 - NumPy
 - Pandas

Social Links

[GitHub](#)

[LinkedIn](#)

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